

Model	Variables	Results						
			24-Hour-Ahead Load Profile Forecasting					
			MAPE (%)	MAE (kWh)	RMSE (kWh)	R2	Params (k)	
MLP	y		9.12 ± 0.14	18.69 ± 0.32	31.74 ± 0.40	0.869	172.72	
	y + H		9.34 ± 0.13	19.06 ± 0.18	32.22 ± 0.34	0.865	124.07	
	y + DT		7.31 ± 0.04	15.93 ± 0.08	24.40 ± 0.21	0.922	126.70	
	y + AC		9.46 ± 0.29	19.56 ± 0.72	32.74 ± 0.87	0.860	349.59	
	y + H + DT		7.35 ± 0.07	15.99 ± 0.15	24.59 ± 0.36	0.921	144.99	
	y + H + AC		9.28 ± 0.20	19.05 ± 0.35	32.40 ± 0.70	0.863	137.82	
	y + DT + AC		7.26 ± 0.09	15.83 ± 0.18	23.51 ± 0.39	0.928	238.70	
	y + H + DT + AC		7.72 ± 0.33	16.95 ± 0.78	25.30 ± 1.08	0.917	228.95	
	y + Profile-1		6.86 ± 0.09	15.02 ± 0.17	21.72 ± 0.28	0.939	86.59	
	y + Profile-2		6.93 ± 0.08	15.09 ± 0.13	21.83 ± 0.23	0.938	240.23	
NBEATSx	y		9.05 ± 0.14	18.59 ± 0.28	31.51 ± 0.34	0.871	2772.26	
	y + H		9.25 ± 0.22	19.04 ± 0.47	31.24 ± 0.50	0.873	3945.39	
	y + DT		7.40 ± 0.16	16.22 ± 0.36	24.26 ± 0.43	0.923	4041.00	
	y + AC		8.92 ± 0.20	18.40 ± 0.33	30.84 ± 0.57	0.876	4041.00	
	y + H + DT		7.75 ± 0.19	16.88 ± 0.34	24.99 ± 0.49	0.919	4283.70	
	y + H + AC		9.25 ± 0.20	19.06 ± 0.30	31.77 ± 0.43	0.869	3369.89	
	y + DT + AC		7.31 ± 0.16	15.97 ± 0.29	23.65 ± 0.43	0.927	3955.94	
	y + H + DT + AC		7.53 ± 0.24	16.4 ± 0.49	24.29 ± 0.63	0.923	4418.70	
	y + Profile-1		7.03 ± 0.09	15.32 ± 0.14	21.89 ± 0.21	0.938	2812.52	
	y + Profile-2		6.96 ± 0.10	15.20 ± 0.15	22.41 ± 0.19	0.935	3107.04	
NHITS	y		9.08 ± 0.17	18.49 ± 0.24	31.72 ± 0.40	0.869	2817.57	
	y + H		9.04 ± 0.24	18.57 ± 0.35	31.50 ± 0.49	0.871	3702.88	
	y + DT		7.34 ± 0.12	16.03 ± 0.25	24.05 ± 0.35	0.925	3014.18	
	y + AC		9.09 ± 0.32	18.64 ± 0.55	30.84 ± 0.88	0.876	3100.27	
	y + H + DT		7.38 ± 0.11	15.96 ± 0.17	24.07 ± 0.30	0.925	3100.12	
	y + H + AC		9.10 ± 0.23	18.94 ± 0.38	31.15 ± 0.52	0.874	4206.77	
	y + DT + AC		7.52 ± 0.13	16.33 ± 0.24	24.66 ± 0.34	0.921	3321.45	
	y + H + DT + AC		7.53 ± 0.26	16.38 ± 0.42	24.19 ± 0.55	0.924	3136.92	
	y + Profile-1		6.96 ± 0.15	15.23 ± 0.27	21.77 ± 0.37	0.938	3272.45	
	y + Profile-2		6.87 ± 0.10	15.09 ± 0.17	22.07 ± 0.22	0.937	3014.18	
TiDE	y		10.83 ± 0.05	22.47 ± 0.06	37.50 ± 0.09	0.817	1335.14	
	y + H		10.81 ± 0.05	22.43 ± 0.07	37.50 ± 0.09	0.817	4517.37	
	y + DT		10.81 ± 0.05	22.43 ± 0.07	37.50 ± 0.09	0.817	4517.37	
	y + AC		10.83 ± 0.05	22.47 ± 0.07	37.40 ± 0.10	0.818	3665.60	
	y + H + DT		10.82 ± 0.04	22.45 ± 0.06	37.49 ± 0.09	0.817	2764.67	
	y + H + AC		10.82 ± 0.04	22.45 ± 0.06	37.49 ± 0.09	0.817	2764.67	
	y + DT + AC		10.82 ± 0.04	22.45 ± 0.06	37.49 ± 0.09	0.817	2764.67	
	y + H + DT + AC		10.81 ± 0.04	22.45 ± 0.07	37.50 ± 0.10	0.817	3368.08	
	y + Profile-1		10.81 ± 0.05	22.43 ± 0.07	37.50 ± 0.09	0.817	4517.37	
	y + Profile-2		10.83 ± 0.05	22.47 ± 0.07	37.40 ± 0.10	0.818	3665.60	
KAN	y		9.30 ± 0.24	18.66 ± 0.34	32.33 ± 0.65	0.864	35773.60	
	y + H		9.09 ± 0.25	18.4 ± 0.38	31.82 ± 0.52	0.868	10180.98	
	y + DT		7.33 ± 0.08	15.91 ± 0.11	24.65 ± 0.28	0.921	850.08	
	y + AC		9.07 ± 0.19	18.18 ± 0.25	31.03 ± 0.54	0.875	24903.68	
	y + H + DT		7.34 ± 0.04	15.88 ± 0.08	24.73 ± 0.17	0.920	1926.81	
	y + H + AC		9.03 ± 0.12	18.24 ± 0.16	30.93 ± 0.27	0.875	1261.44	
	y + DT + AC		7.23 ± 0.16	15.78 ± 0.35	23.65 ± 0.66	0.927	0.00	
	y + H + DT + AC		7.36 ± 0.08	15.86 ± 0.14	23.78 ± 0.24	0.926	4017.60	
	y + Profile-1		7.01 ± 0.13	15.35 ± 0.24	21.89 ± 0.34	0.938	17600.66	
	y + Profile-2		6.77 ± 0.09	14.78 ± 0.17	21.41 ± 0.24	0.940	7027.36	
TCN	y		12.77 ± 6.13	26.31 ± 14.18	40.07 ± 16.24	0.757	1558.23	
	y + H		9.32 ± 0.24	18.87 ± 0.43	32.05 ± 0.59	0.866	703.39	
	y + DT		7.77 ± 0.19	16.77 ± 0.39	25.96 ± 0.56	0.912	204.72	
	y + AC		9.91 ± 0.54	20.18 ± 1.02	33.06 ± 1.23	0.857	4077.16	
	y + H + DT		7.60 ± 0.15	16.39 ± 0.28	25.39 ± 0.49	0.916	254.61	
	y + H + AC		9.98 ± 3.05	20.11 ± 4.91	32.90 ± 6.59	0.853	1938.64	
	y + DT + AC		7.60 ± 0.15	16.39 ± 0.28	25.39 ± 0.49	0.916	254.61	
	y + H + DT + AC		7.94 ± 0.64	17.12 ± 1.31	26.34 ± 4.09	0.907	4092.06	
	y + Profile-1		7.18 ± 0.15	15.53 ± 0.30	22.64 ± 0.41	0.933	132.13	
	y + Profile-2		7.11 ± 0.15	15.43 ± 0.29	22.50 ± 0.36	0.934	132.11	
RNN	y		13.48 ± 1.29	27.48 ± 3.11	46.13 ± 4.52	0.720	225.09	
	y + H		16.53 ± 1.87	31.73 ± 3.04	53.22 ± 5.21	0.628	452.67	
	y + DT		8.51 ± 0.70	18.5 ± 1.57	28.91 ± 2.69	0.890	84.80	
	y + AC		20.15 ± 1.18	39.94 ± 2.84	62.77 ± 3.61	0.485	292.71	
	y + H + DT		8.27 ± 0.08	17.9 ± 0.13	27.82 ± 0.14	0.899	66.87	
	y + H + AC		15.61 ± 1.70	30.99 ± 3.30	50.41 ± 4.57	0.666	252.96	
	y + DT + AC		8.62 ± 0.28	18.68 ± 0.55	28.36 ± 0.78	0.895	74.00	
	y + H + DT + AC (1)		-	-	-	-	-	
	y + Profile-1		7.73 ± 0.10	16.68 ± 0.19	24.07 ± 0.19	0.925	60.67	
	y + Profile-2		7.35 ± 0.05	15.84 ± 0.07	23.03 ± 0.13	0.931	1.03	
GRU	y		22.64 ± 6.46	48.61 ± 13.82	70.86 ± 13.24	0.323	741.97	
	y + H		11.82 ± 2.50	23.39 ± 4.86	39.10 ± 6.49	0.795	120.61	
	y + DT		8.61 ± 0.74	18.75 ± 1.58	29.33 ± 2.67	0.887	113.66	
	y + AC		14.7 ± 4.41	30.66 ± 9.54	49.71 ± 12.87	0.657	167.50	
	y + H + DT		8.28 ± 0.25	17.89 ± 0.49	27.28 ± 0.76	0.903	121.40	
	y + H + AC		10.41 ± 0.69	21.22 ± 1.14	35.51 ± 1.95	0.835	449.01	
	y + DT + AC		8.92 ± 0.88	19.23 ± 1.89	29.42 ± 3.35	0.886	215.99	
	y + H + DT + AC		13.13 ± 8.41	29.01 ± 19.49	40.71 ± 24.09	0.709	198.00	
	y + Profile-1		7.79 ± 0.18	16.80 ± 0.28	24.22 ± 0.31	0.924	247.33	
	y + Profile-2		7.45 ± 0.16	16.07 ± 0.28	22.93 ± 0.34	0.932	208.68	
LSTM	y		12.63 ± 2.57	26.25 ± 5.62	43.27 ± 7.28	0.749	1194.09	
	y + H		12.54 ± 2.10	24.83 ± 3.74	41.81 ± 5.88	0.768	487.05	
	y + DT		8.32 ± 0.35	18.04 ± 0.69	28.32 ± 1.03	0.895	839.26	
	y + AC		17.98 ± 7.99	39.49 ± 20.06	57.46 ± 21.77	0.508	1124.15	
	y + H + DT		8.25 ± 0.11	17.86 ± 0.18	27.82 ± 0.21	0.899	75.00	
	y + H + AC		13.68 ± 3.64	27.54 ± 7.24	44.59 ± 9.27	0.730	696.81	
	y + DT + AC		8.67 ± 0.78	18.84 ± 1.74	31.47 ± 13.22	0.848	179.04	
	y + H + DT + AC		8.52 ± 0.57	18.40 ± 1.32	29.64 ± 13.81	0.861	470.73	
	y + Profile-1		7.79 ± 0.14	16.78 ± 0.23	24.17 ± 0.25	0.924	18.65	
	y + Profile-2		7.38 ± 0.09	15.89 ± 0.13	23.02 ± 0.15	0.931	13.91	
TFT (2)	y		-	-	-	-	-	
	y + H		-	-	-	-	-	
	y + DT		-	-	-	-	-	
	y + AC		-	-	-	-	-	
	y + H + DT		-	-	-	-	-	
	y + H + AC		-	-	-	-	-	
	y + DT + AC		-	-	-	-	-	
	y + H + DT + AC		-	-	-	-	-	
	y + Profile-1		-	-	-	-	-	
	y + Profile-2		-	-	-	-	-	
PatchTST (3)	y		9.47 ± 0.19	19.43 ± 0.48	32.2 ± 0.71	0.865	180.12	
RMoK (4)	y		10.51	21.76	35.98	0.831	4.78	

(1) The results for this specific experiment are not included considering errors of significant magnitude (e.g., the result for MAPE was over 4000%)

(2) Due to limitations concerning computational resources the experiments for 24-hour-ahead load profile forecasts were not possible for TFT

(3) The PatchTST model, of the NeuralForecast library, didn't support exogenous variables at the time of the study

(4) The RMoK model, of the NeuralForecast library, didn't support the inclusion of exogenous variables at the time of the study, the variation of the seed doesn't affect the outputs of this model (for the same configuration)